

A table representing gasoline prices per gallon at one gas station is shown.

Gas (gallons)	Price (dollars)
$\frac{1}{2}$	\$1.49
10	\$29.80
3	\$8.94
17	\$50.66

Use the table for questions 1 through 5.

1. A car that holds 20 gallons of gas will pay \$59.60.

$$1.49 \cdot 2 = 2.98/\text{gallon} \quad \frac{59.60}{2.98} = 20 \text{ gallons}$$

2. A push mower typically uses 1 gallon of gas, which will cost \$2.98 to fill up the tank.

$$\frac{.5 \text{ gallon}}{\$1.49} \times \frac{2}{2} = \frac{1 \text{ gallon}}{\$2.98}$$

3. The road trip from Orlando, FL to Chicago, IL was 1152 miles. If the automobile got 24 miles per gallon on the trip, ☐ 42 gallons of gas was needed at a

cost of ☐ \$125.16
☐ \$143.04 .

$$\frac{1152}{24} = 48$$

$$48 \cdot 2.98 = \$143.04$$

4. One unit, or gallon, of gas to its cost can be written as 1: \$2.98.

5. The gas station down the street has a unit rate of $\frac{1}{\$3.09}$. That station has

an ☐ increased
☐ decreased unit rate than the station represented by the table.

The Shine Central Car Wash has an Ultimate Car Wash that uses 3.5L of soap and 2L of wax for every wash.

Soap	Wax	Car Washes
42L	24L	12
147L	84L	42
31.5L	18L	9

6. Complete the table that represents the amount of soap and wax used for the Ultimate Car Wash.
7. What is the unit rate of wax to every car wash?
2L per wash
8. The ratio of soap to wax for 12 car washes is 42:24.
9. The liters of soap needed for a day of

☐ 53

☐ 26

 Ultimate Car Washes is

☐ 185.5

☐ 52
10. A delivery of 460 liters of wax was just made to the Shine Central Car Wash. That is enough wax for 230 ultimate car washes. They will need

☐ 805

☐ 1104

 liters of soap to match that number of washes.